Caron 2020 Acute Lymphoblastic Leukemia — analysis summary

RUN 2 OF 4

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony(θ=10) lifts per-neighbourhood iLISI 1.0→4.7 of 8 samples; the 3 genetic subtypes intermix on one connected manifold (single component after island removal). Read on the embedding + iLISI sweep, not a scalar.

Gate 2 — module geography → GO (non-branching)

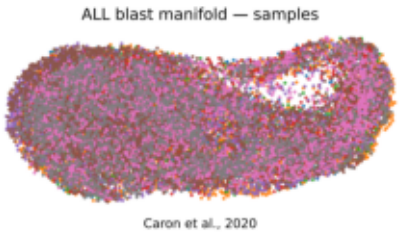
PPT tree = 0 forks → non-branching. Organized Gast↔pSom cross-repression (r=−0.45; −0.37 after removing cell cycle) + Crest inverse gradient (r=−0.72 vs pSom) SURVIVE cell-cycle regression → real developmental axis, not a proliferation artefact. FLAG: Crest leans to the Gast end (inversion of ideal pSom-arm), passes anyway.

Gate 3 — hub cross-repression → NA

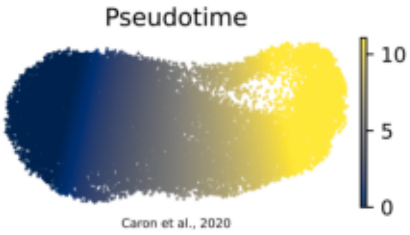
Non-branching topology has no bifurcation to test (branching-only gate).

STEP FIGURES

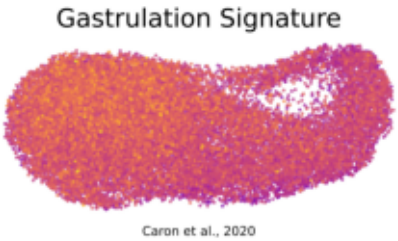
Manifold — samples



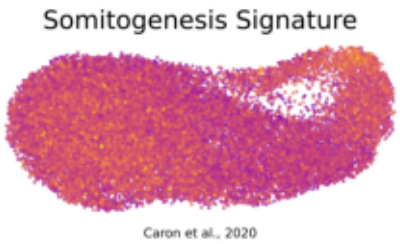
Pseudotime



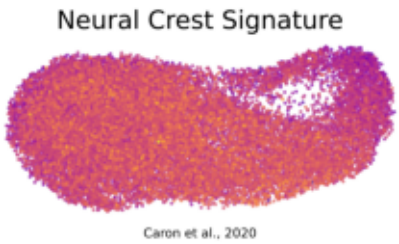
Gastrulation



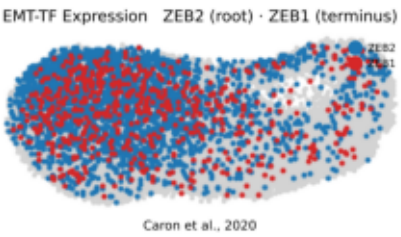
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Pediatric B/T-cell ALL (Caron et al., 2020), 10x — 25,788 malignant blasts, 8 samples, 3 genetic subtypes (ETV6-RUNX1, HHD, pre-T).

Verdict.
Pattern reproduces as a non-branching developmental axis — organized Gast↔pSom cross-repression with a terminal Crest gradient, cell-cycle-independent. Gate 1 GO / Gate 2 GO / Gate 3 NA. Crest-arm inversion flagged. Caveat: the dominant axis is the CELL CYCLE (pseudotime ρ=+0.69 with cycling; cc explains R^2=0.37/0.17/0.11 of Gast/pSom/Crest); the developmental cross-repression is the residual structure surviving cell-cycle removal (Zhou 2020 precedent). EMT-TFs ZEB2 (root) → ZEB1 (terminus); other factors <3% detection in blasts.

Parameter choices

Threshold	none — 8 samples span 1,585–5,816 cells with no clean multiplicative cliff; gap rule is a no-op (keep all)	**Embedding**	n_pcs=20, UMAP spread=1.3; Harmony θ=10 (iLISI plateaus by θ=5; subtypes need strong integration)
Geometry	NON-BRANCHING principal curve (ElPiGraph, fold-free self-approach 3.0); PPT tree gave 0 branch nodes	**Root**	quiescent Gast/pSom-high pole (origin; Crest is late)
Cell cycle	dominant axis is cell cycle (pt ρ=+0.69); cc explains R^2=0.37/0.17/0.11 of Gast/pSom/Crest; dev cross-repression survives cc removal	**EMT-TFs**	ZEB2 (root) → ZEB1 (terminus); other factors <3% detection